

Cognitive Biases in Spiral-of-Silence Opinion Models: Analysis and Large-Scale Agent-Based Simulation

Frank Valencia^{1,2}, Juan Fco. Díaz³, Jesús Aranda³,
David Gaona³, Juan Marcos Caicedo^{3*}

¹Pontificia Universidad Javeriana Cali, Cali, Colombia

²CNRS LIX, École Polytechnique de Paris, Palaiseau, France

³Universidad del Valle, Escuela de Ingeniería de Sistemas y Computación,
Cali, Colombia

frank.valencia@gmail.com,

{juanfco.diaz, jesus.aranda, david.gaona, juan.marcos.caicedo}@correounivalle.edu.co

Abstract

We introduce two DeGroot-based models of opinion formation in social networks that combine cognitive biases with the Spiral of Silence: CBSOM⁻ (memoryless) and CBSOM⁺ (memory-based). A community of agents is represented as a weighted directed influence graph. At each step, every agent passes its disagreement with each influencer through a bias function (confirmation, backfire, authority, or insular) and updates its opinion accordingly. It then decides whether to speak or to remain silent. Silent agents exert no influence in CBSOM⁻, whereas in CBSOM⁺ they keep influencing others through their last public opinion. We prove that two properties of the memoryless base model are preserved under every continuous receptive-resistant bias: global silence never lasts more than one round, and the extreme opinions contract monotonically. For the scaled linear biases $f_\alpha(x) = \alpha x$ with $\alpha \in (0, 1)$, we prove consensus in cliques of every size $n \geq 2$. For $n = 2$, this removes the period-2 oscillation of the base model. Simulations on networks of up to 10^6 agents show that confirmation bias preserves consensus under memoryless dynamics, whereas backfire and authority biases almost always prevent it. Memory suppresses consensus even when most edges carry confirmation bias.

Keywords: Opinion dynamics, Spiral of Silence, DeGroot model, Cognitive biases, Consensus, Polarization, Agent-based simulation, Social networks.

1 Introduction

Social networks have a strong impact on *opinion formation*, often resulting in polarization. Individuals express their views, are exposed to the views of others, and may revise their own beliefs in response. Whether this feedback loop ends in broad consensus or in entrenched polarization depends on the structure of the network and on the behavioral rules that govern each interaction. Formal models of this process allow us to characterize precisely the conditions under which each outcome arises.

The DeGroot model [1] is one of the most prominent formalisms for social learning and opinion formation [2]. A community is represented as a weighted directed graph, the *influence graph*, whose nodes are *agents* and whose edges encode how strongly agents influence one another. Each agent holds an opinion in $[0, 1]$, which indicates the strength of its agreement with a given proposition. Agents repeatedly revise their opinions by taking a weighted average of the opinions of those who influence them. A fundamental theoretical result guarantees convergence to *consensus* whenever the influence graph is strongly connected and agents retain non-zero self-influence [1, 2]. The significance of this result lies in the fact that consensus is a central problem in social learning. Indeed, the inability to reach consensus is a sign of a polarized community.

Nevertheless, the DeGroot model makes two assumptions that can be overly constraining in real social networks. First, it assumes that agents *always* express their opinions, disregarding the social pressure that

*Corresponding author.

may lead individuals to withhold their views. Second, it assumes that agents *linearly* assimilate the opinions of those who influence them, ignoring the nonlinear psychological responses that shape how people react to disagreement. Two well-documented phenomena challenge these assumptions: the *Spiral of Silence* and *cognitive biases*, respectively.

The Spiral of Silence, introduced by Noelle-Neumann [3, 4], posits that individuals who perceive their opinion to be in the minority tend to *self-censor*. They withhold expression to avoid social isolation, and their silence reinforces the dominance of the majority view. Empirical studies confirm that this mechanism is active both offline and online [5–9]. Simulation studies show that it can substantially distort the collective opinion landscape in digitally mediated networks [9–12]. Aranda et al. [13] recently gave a formal treatment of this mechanism: they extended the DeGroot model with silence and derived conditions under which consensus forms or fails. That work is the direct predecessor of the present paper.

Cognitive biases add a further source of nonlinearity. Individuals do not respond neutrally to information that conflicts with their beliefs. *Confirmation bias* leads them to discount disagreeing views; the *backfire effect* makes them harden their positions when confronted with contrary evidence; *authority bias* amplifies the influence of perceived high-status sources; and *insular bias* makes them ignore external opinions altogether [14–17]. These distortions change how agents assimilate and reject information, and they can foster polarization in ways that the DeGroot model cannot capture. Alvim et al. [17] encoded these four biases as nonlinear *bias functions* applied to opinion disagreements, and derived conditions under which consensus is preserved or fails in the synchronous DeGroot model. Their framework does not consider silence. Conversely, the model of Aranda et al. [13] considers silence but not cognitive biases.

In this paper, we combine both phenomena in two *Cognitive-Bias Silence Opinion Models* (CBSOM). The state of the system records, for each agent, an opinion in $[0, 1]$ and whether the agent is *vocal* or *silent*. At each time unit, every agent performs two actions. First, it revises its opinion: the disagreement with each influencer passes through a bias function, which determines how much of that disagreement the agent actually absorbs. Second, it decides whether to speak in the next round: it remains vocal only if enough of its influencers hold opinions close to its own, within a *tolerance radius*. Silent agents still revise their opinions; they only stop expressing them. The two models differ in what silent agents contribute. In the memoryless model CBSOM[−], silent agents exert no influence. In the memory-based model CBSOM⁺, they keep influencing others through the last opinion that they expressed.

We focus on the problem of convergence to consensus. We first show that the bias layer does not interfere with two structural properties of the memoryless base model SOM[−] [13]: global silence never lasts more than one round, and the maximum and minimum opinions contract monotonically. Both properties hold for every continuous bias in the *receptive-resistant* region of [17], i.e., for biases that move an agent towards an influencer without overshooting. We then restrict our attention to the *scaled linear* biases $f_\alpha(x) = \alpha x$ with $\alpha \in (0, 1)$, and prove consensus on cliques of every size $n \geq 2$. The case $n = 2$ is of particular interest: on two-agent cliques, SOM[−] can oscillate forever between two opinions, and every scaled linear bias removes this oscillation. Finally, we use large-scale simulations, with up to 10^6 agents, to study the settings that our formal analysis does not yet cover: non-clique graphs, biases outside the receptive-resistant region, mixtures of biases, and memory.

To the best of our knowledge, this is the first work that embeds cognitive bias functions into a Spiral-of-Silence model and gives formal consensus guarantees for the resulting models.

Contributions. The contributions of this paper are the following:

1. We introduce CBSOM[−] and CBSOM⁺, two Cognitive-Bias Silence Opinion Models that couple confirmation, backfire, authority, and insular bias functions with memoryless and memory-based update rules, respectively (Section 2).
2. We prove that two structural properties of the base model SOM[−] [13], namely the impossibility of perpetual global silence and the monotonic contraction of extreme opinions, are preserved under every continuous bias function in the receptive-resistant region (Section 3).
3. For the scaled linear biases $\mathcal{F}_\alpha = \{f_\alpha(x) = \alpha x \mid \alpha \in (0, 1)\}$, we prove consensus in cliques for all $n \geq 2$. This removes the period-2 oscillation that prevents consensus on two-agent cliques in the base model (Section 3).
4. Through simulations on uniform and non-uniform networks with up to 10^6 agents, we show that (i) confirmation bias, which lies in the receptive-resistant region, largely preserves consensus; (ii) backfire and authority biases prevent consensus even in dense graphs; (iii) the density needed for consensus grows only sub-linearly with network size; and (iv) memory suppresses consensus almost entirely as networks grow (Section 4).

Organization. Section 2 recalls the DeGroot model, the bias functions of [17], and the Silence Opinion Models of [13], and then defines CBSOM[−] and CBSOM⁺. Section 3 presents the formal analysis. Section 4 reports the simulation study. Section 5 discusses related work and concludes. For the sake of readability, some proofs are

only outlined in the main text; the complete proofs are given in Appendix A. The simulator that supports our experiments is publicly available at https://github.com/juanmarcosdev/belief_evolution_simulator.

2 Models

This section introduces the three building blocks of the proposed models—the DeGroot model [1], the cognitive bias framework [17], and the Spiral-of-Silence opinion models [13]—and defines CBSOM[−] and CBSOM⁺, the Cognitive-Bias Silence Opinion Models that combine all three.

2.1 The DeGroot Model

The DeGroot model [1] formalizes social learning as repeated weighted averaging over a fixed *influence graph*.

Definition 1 (Influence Graph). An *influence graph* is a weighted directed graph $G = (A, E, I)$ with $A = \{1, \dots, n\}$, $n > 1$, the set of *agents*; $E \subseteq A \times A$ the set of directed edges; and $I : A \times A \rightarrow [0, 1]$ the weight function satisfying $I(i, j) = 0 \iff (i, j) \notin E$ and, for each $i \in A$,

$$\sum_{j \in N(i) \cup \{i\}} I(j, i) = 1, \quad (1)$$

where $N(i) = \{j \in A \setminus \{i\} : (j, i) \in E\}$ is the set of *influencers* of i .

The vertices in A represent the n agents of a community. An edge $(j, i) \in E$ means that agent j influences agent i , and the weight $I(j, i)$ denotes the strength of that influence: the higher the value, the stronger the influence. Equation (1) requires that the weights that agent i gives to its influencers and to itself add up to 1.

Each agent i holds an *opinion* $B_i^t \in [0, 1]$ at time t , interpreted as the degree of agreement with an underlying proposition (e.g., “*AI poses a threat to humanity*”). If $B_i^t = 0$, agent i completely disagrees with the proposition; if $B_i^t = 1$, it completely agrees with it. The full opinion vector is $\mathbf{B}^t \in [0, 1]^n$. The synchronous DeGroot update reads:

$$B_i^{t+1} = \sum_{j=1}^n I_{ij} B_j^t, \quad \text{equivalently} \quad \mathbf{B}^{t+1} = I \mathbf{B}^t, \quad (2)$$

where $I = [I_{ij}]$ is row-stochastic.

Theorem 1 (DeGroot Consensus [1]). *If I is row-stochastic and primitive (i.e., $\exists k$ such that all entries of I^k are positive), then there exists $\mathbf{B}^*$ with identical components such that $\lim_{t \rightarrow \infty} \mathbf{B}^t = \mathbf{B}^*$.*

Despite its tractability, the DeGroot model makes two assumptions challenged by real social dynamics: it assumes that agents always express their opinions freely (no strategic silence), and it assumes *linear* assimilation of neighbors’ opinions, ignoring nonlinear psychological responses to disagreement [4, 17]. Sections 2.2 and 2.3 address each limitation in turn, before they are combined in Sections 2.4–2.5.

2.2 Cognitive Bias Functions

Following [17], we model cognitive bias as a function that nonlinearly transforms the opinion disagreement between an agent and each of its influencers before the update is applied.

Definition 2 (Bias Function [17]). A *bias function* for agent i with respect to agent j is a continuous map $\beta_{i,j} : [-1, 1] \rightarrow [-1, 1]$ that transforms the raw disagreement $x = B_j^t - B_i^t$.

Intuitively, $\beta_{i,j}(x)$ is the part of the disagreement x that agent i actually absorbs from agent j . The identity function $\beta_{i,j}(x) = x$ corresponds to the linear response of the DeGroot model. The bias-extended DeGroot update replaces the linear term $(B_j - B_i)$ with $\beta_{i,j}(B_j - B_i)$:

$$B_i^{t+1} = B_i^t + \sum_{j \in N(i)} I_{ji} \beta_{i,j}(B_j^t - B_i^t). \quad (3)$$

The original formulation of [17] includes an explicit $[\cdot]_0^1$ clamp to enforce $B_i^{t+1} \in [0, 1]$. Our formal results assume bias functions in the Receptive-Resistant region R defined below. In that region the update stays within $[0, 1]$ without clamping (Lemma 7), so we omit the clamp from here on. The four representative bias functions examined in this paper are:

- **Confirmation bias** (*conf*): agents attenuate distant disagreements, $\text{conf}(x) = x(1 + \delta - |x|)/(1 + \delta)$, $\delta > 0$ [15, 17].
- **Backfire effect** (*backf*): large disagreements trigger repulsion, $\text{backf}(x) = -x^3$ [16, 17].
- **Authority bias** (*fan*): agents follow perceived authorities with maximal response, $\text{fan}(x) = x/|x|$ for $x \neq 0$, 0 otherwise [14, 17].
- **Insular bias** (*ins*): agents ignore all external influence, $\text{ins}(x) = 0$ [17].

Bias regions. Alvim et al. [17] classify bias functions by the region of the square $[-1, 1]^2$ in which the graph of $(x, \beta(x))$ lies:

- *Receptive-Resistant* (R): $0 < |\beta(x)| < |x|$; confirmation bias belongs to this region. In synchronous DeGroot dynamics, consensus is preserved on strongly connected graphs under continuous R -biases [17].
- *Malleability* (M): $|\beta(x)| \geq |x|$; authority bias is the extreme case.
- *Backfire*: sign reversal, $\text{sign}(\beta(x)) \neq \text{sign}(x)$; consensus typically fails.
- *Insular* (I): zero response, $\beta(x) = 0$.

The region R is central to this paper. A bias in R moves an agent towards each influencer, but never beyond it. Together with continuity, this property is the key analytical lever for extending consensus guarantees to the models introduced below.

2.3 Spiral-of-Silence Opinion Models

The Spiral of Silence [3, 4] posits that agents perceiving themselves as a minority tend to self-censor to avoid social isolation. Aranda et al. [13] formalize this mechanism as an augmentation of the DeGroot framework.

Definition 3 (Silence Opinion Model [13]). A *Silence Opinion (SO) model* is a quadruple $(G, \mathbf{B}^0, \mathbf{S}^0, \mu_G)$, where $G = (A, E, I)$ is an influence graph, $\mathbf{B}^0 \in [0, 1]^n$ is the initial opinion vector, $\mathbf{S}^0 \in \{0, 1\}^n$ is the initial silence vector ($S_i^t = 1$ meaning agent i is *vocal* at time t), and $\mu_G : [0, 1]^n \times \{0, 1\}^n \times \mathbb{N} \rightarrow [0, 1]^n \times \{0, 1\}^n$ is the joint update function.

Definition 4 (Consensus). Agents in an SO model *reach consensus* if $\exists v \in [0, 1]$ such that $\lim_{t \rightarrow \infty} B_i^t = v$ for all $i \in A$.

2.3.1 Memoryless Model SOM^-

In SOM^- , silent agents contribute nothing to their neighbors' updates [13]. Writing $x \sim_{\tau_i} y$ iff $|x - y| \leq \tau_i$ for tolerance radius $\tau_i \in [0, 1]$:

$$B_i^{t+1} = B_i^t + \sum_{j \in N(i)} I_{ji} \cdot S_j^t \cdot (B_j^t - B_i^t), \quad (4)$$

$$S_i^{t+1} = \begin{cases} 1 & \text{if } \lceil |N(i)|/2 \rceil \leq |\{j \in N(i) \mid B_i^t \sim_{\tau_i} B_j^t\}|, \\ 0 & \text{otherwise,} \end{cases} \quad (5)$$

where $N^t(i) = \{j \in N(i) : S_j^t = 1\}$. Intuitively, agent i goes silent when a majority of its currently vocal neighbors hold opinions too far from its own. Silent agents still revise their own opinions through (4); they only stop expressing them.

2.3.2 Memory-Based Model SOM^+

In SOM^+ , silent agents retain influence through the last opinion they publicly expressed [13]. Assuming $\mathbf{S}^0 = \mathbf{1}_n$, let $t'_j = \max\{u \leq t : S_j^u = 1\}$ and $\hat{B}_j^t = B_j^{t'_j}$ (*last public opinion* of agent j):

$$B_i^{t+1} = B_i^t + \sum_{j \in N(i)} I_{ji} \cdot (\hat{B}_j^t - B_i^t), \quad (6)$$

$$S_i^{t+1} = \begin{cases} 1 & \text{if } \lceil |N(i)|/2 \rceil \leq |\{j \in N(i) \mid B_i^t \sim_{\tau_i} \hat{B}_j^t\}|, \\ 0 & \text{otherwise.} \end{cases} \quad (7)$$

Unlike SOM^- , the model SOM^+ is non-Markovian: its evolution depends on the history of silence events through $\hat{B}_j^t$.

2.3.3 Key Results for SOM^-

The following results from [13] underpin the analysis in Section 3.

Lemma 2 (Bounded opinions [13]). *For every SOM^- model and all t : $\min(\mathbf{B}^t) \leq B_i^{t+1} \leq \max(\mathbf{B}^t)$ for all $i \in A$.*

Corollary 3 (Monotone extremes [13]). *In every SOM^- model, $\max(\mathbf{B}^t)$ is non-increasing and $\min(\mathbf{B}^t)$ is non-decreasing in t .*

Theorem 4 (Consensus in SOM^- cliques [13]). *Let $(G, \mathbf{B}^0, \mathbf{S}^0, \mu_G)$ be a SOM^- model on an n -agent clique with $n \geq 3$. Then all agents converge to consensus.*

2.4 CBSOM $^-$: Memoryless with Cognitive Biases

Definition 5 (CBSOM $^-$). An SO model $(G, \mathbf{B}^0, \mathbf{S}^0, \mu_G)$ is *memoryless with cognitive biases* (CBSOM $^-$) if, for each $i \in A$ and $t \in \mathbb{N}$:

$$B_i^{t+1} = B_i^t + \sum_{j \in N(i)} I_{ji} \cdot S_j^t \cdot \beta_{i,j}(B_j^t - B_i^t), \quad (8)$$

$$S_i^{t+1} = \begin{cases} 1 & \text{if } \lceil |N^t(i)|/2 \rceil \leq |\{j \in N^t(i) \mid B_i^t \sim_{\tau_i} B_j^t\}|, \\ 0 & \text{otherwise,} \end{cases} \quad (9)$$

where each $\beta_{i,j} : [-1, 1] \rightarrow [-1, 1]$ is a bias function.

Equation (8) extends (4) by replacing the linear disagreement term $(B_j^t - B_i^t)$ with the biased term $\beta_{i,j}(B_j^t - B_i^t)$. The silence factor S_j^t continues to zero out silent agents' contributions entirely; thus, *bias determines how vocal neighbors are processed, while silence determines which neighbors are visible*. The two mechanisms remain structurally decoupled. The following example illustrates both.

Example 1. Consider a clique of three agents in which each agent gives weight $1/2$ to each of the other two; that is, $I_{ji} = 1/2$ for $j \neq i$ and $I_{ii} = 0$. Let $\mathbf{B}^0 = (1, 0.5, 0)$, $\mathbf{S}^0 = (1, 1, 1)$, and $\tau_i = 0.4$ for every agent, and let every edge carry the bias $\beta_{i,j}(x) = x/2$, which halves each disagreement. By (8), agent 1 updates its opinion to

$$B_1^1 = 1 + \frac{1}{2} \cdot \frac{1}{2} (0.5 - 1) + \frac{1}{2} \cdot \frac{1}{2} (0 - 1) = 0.625,$$

and similarly $\mathbf{B}^1 = (0.625, 0.5, 0.375)$. Without bias ($\beta_{i,j}(x) = x$) we would obtain $(0.25, 0.5, 0.75)$ instead: agents 1 and 3 would swap sides, whereas the biased agents move only part of the way towards each other.

Now consider silence. At $t = 0$, every agent disagrees with both of its neighbors by more than $\tau_i = 0.4$, so by (9) all agents fall silent: $\mathbf{S}^1 = (0, 0, 0)$. With no vocal agent, no opinion changes, and $\mathbf{B}^2 = \mathbf{B}^1$. At $t = 1$ no agent hears any disagreement, so all agents speak again: $\mathbf{S}^2 = (1, 1, 1)$. From then on all pairwise distances are at most $0.25 < \tau_i$, so the agents remain vocal and approach the consensus value 0.5. This episode of global silence is not a coincidence: Lemma 5 in Section 3 shows that global silence never lasts more than one round.

Remark 2.1 (DeGroot as a special case). Setting $\tau_i = 1$, $\mathbf{S}^0 = \mathbf{1}_n$, and $\beta_{i,j}(x) = x$ recovers the classical DeGroot update from CBSOM $^-$.

2.5 CBSOM $^+$: Memory-Based with Cognitive Biases

Definition 6 (CBSOM $^+$). An SO model $(G, \mathbf{B}^0, \mathbf{S}^0, \mu_G)$ is *memory-based with cognitive biases* (CBSOM $^+$) if $\mathbf{S}^0 = \mathbf{1}_n$ and, for each $i \in A$ and $t \in \mathbb{N}$:

$$B_i^{t+1} = B_i^t + \sum_{j \in N(i)} I_{ji} \cdot \beta_{i,j}(\hat{B}_j^t - B_i^t), \quad (10)$$

$$S_i^{t+1} = \begin{cases} 1 & \text{if } \lceil |N(i)|/2 \rceil \leq |\{j \in N(i) \mid B_i^t \sim_{\tau_i} \hat{B}_j^t\}|, \\ 0 & \text{otherwise,} \end{cases} \quad (11)$$

where $\hat{B}_j^t = B_j^{t'}$ with $t' = \max\{u \leq t : S_j^u = 1\}$.

The model CBSOM $^+$ applies the bias function to historical public opinions: $\beta_{i,j}$ filters how agent i assimilates the last expressed belief of neighbor j , regardless of whether j is currently vocal. This creates a coupling between time and cognition that is absent in CBSOM $^-$. The frozen opinion of a silent agent is processed through the bias of each of its neighbors at every step, until that agent speaks again.

Remark 2.2 (DeGroot as a special case). Setting $\tau_i = 1$ and $\beta_{i,j}(x) = x$ recovers DeGroot from CBSOM $^+$: perpetual vocality ensures $\hat{B}_j^t = B_j^t$ for all j, t , reducing (10) to (2).

2.6 Relationship Among Models

Table 1 summarizes the model hierarchy. The DeGroot model is a special case of both CBSOM⁻ and CBSOM⁺ (Remarks 2.1–2.2). The SOM models are the special cases of the CBSOM models obtained by setting $\beta_{i,j}(x) = x$. Therefore, every result proved for CBSOM with identity biases recovers the corresponding SOM result.

Table 1: Hierarchy of opinion models. $\checkmark$ = mechanism present; $-$ = absent or identity

Model	Silence	Memory	Bias $\beta_{i,j}$
DeGroot [1]	–	–	identity
SOM ⁻ [13]	memoryless	–	identity
SOM ⁺ [13]	memory-based	$\checkmark$	identity
CBSOM ⁻ (<i>this work</i>)	memoryless	–	general
CBSOM ⁺ (<i>this work</i>)	memory-based	$\checkmark$	general

3 Analysis

In this section, we study the formal properties of CBSOM⁻ under cognitive bias functions from the Receptive-Resistant region R . Our main results state that CBSOM⁻ reaches consensus on cliques of every size $n \geq 2$ when all biases are scaled linear functions $f_\alpha(x) = \alpha x$ (Theorems 12 and 14). We outline the argument in three stages:

1. We show that the silence and bias mechanisms do not interfere: global silence never lasts more than one round, and the maximum and minimum opinions converge to some values U and L with $L \leq U$ (Section 3.1). These results hold for every continuous bias in R and every influence graph.
2. We introduce the *scaled linear bias family* $\mathcal{F}_\alpha$ and show that, on cliques, it contracts the opinion spread $\max(\mathbf{B}^t) - \min(\mathbf{B}^t)$ below every positive threshold (Section 3.2).
3. We combine both stages: the spread cannot fall below $U - L$, so $U = L$, and every agent converges to that common value (Section 3.3). The two-agent case requires a separate argument.

3.1 Structural Properties under R -Biases

A central motivation for restricting to the Receptive-Resistant region R is that continuous R -biases guarantee consensus in the synchronous DeGroot setting on strongly connected graphs [17]. We ask whether the silence mechanism of CBSOM⁻ preserves the structural invariants that underpin the SOM⁻ consensus proofs [13]. We show that it does. The silence criterion in Eq. (9) depends only on the raw opinion proximity $|B_i^t - B_j^t| \leq \tau_i$, not on the bias-filtered update. Hence the cognitive bias layer *decouples* from the silence decision.

Lemma 5 (All-silent state lasts at most one round). *Let $(G, \mathbf{B}^0, \mathbf{S}^0, \mu_G)$ be a CBSOM⁻ model. For every $t \in \mathbb{N}$, if $S_i^t = 0$ for all $i \in A$, then $S_i^{t+1} = 1$ for all $i \in A$.*

Proof. If $S_i^t = 0$ for all i , then $N^t(i) = \emptyset$ for every i , so $\lceil |N^t(i)|/2 \rceil = 0$ and $|\{j \in N^t(i) \mid B_i^t \sim_{\tau_i} B_j^t\}| = 0$. The inequality $0 \leq 0$ holds trivially, giving $S_i^{t+1} = 1$. The argument does not use the bias functions $\beta_{i,j}$. $\square$

Corollary 6 (No perpetual global silence in CBSOM⁻). *In every CBSOM⁻ model, for every $t \in \mathbb{N}$ there exists $i \in A$ such that $S_i^t = 1$ or $S_i^{t+1} = 1$.*

Intuitively, silence can block every channel of influence for one round, but not for two consecutive rounds. The next result extends Lemma 2 to CBSOM⁻: no agent can move beyond the current extreme opinions.

Lemma 7 (Bounded opinions under CBSOM⁻). *Let $(G, \mathbf{B}^0, \mathbf{S}^0, \mu_G)$ be a CBSOM⁻ model with continuous bias functions $\beta_{i,j} \in R$ for all edges. For every $t \in \mathbb{N}$ and all $i \in A$:*

$$\min(\mathbf{B}^t) \leq B_i^{t+1} \leq \max(\mathbf{B}^t).$$

Proof sketch. For each vocal neighbor $j \in N^t(i)$, let $x = B_j^t - B_i^t$. Since $\beta_{i,j} \in R$, we have $\text{sign}(\beta_{i,j}(x)) = \text{sign}(x)$ and $|\beta_{i,j}(x)| \leq |x|$. Hence the term $I_{ji} \cdot \beta_{i,j}(x)$ moves B_i^t towards B_j^t without overshooting. The row-stochasticity of I then yields the bound by a convexity argument. The full proof is in Appendix A. $\square$

Corollary 8 (Monotone extremes in CBSOM⁻). *In every CBSOM⁻ model with continuous R -biases, $\max(\mathbf{B}^t)$ is non-increasing and $\min(\mathbf{B}^t)$ is non-decreasing in t .*

Theorem 9 (Limits of extremes in CBSOM⁻). *For every CBSOM⁻ model with continuous R -biases, there exist $U, L \in [0, 1]$ with $L \leq U$ such that $\lim_{t \rightarrow \infty} \max(\mathbf{B}^t) = U$ and $\lim_{t \rightarrow \infty} \min(\mathbf{B}^t) = L$.*

Proof. The sequence $\{\max(\mathbf{B}^t)\}$ is bounded below by 0 and non-increasing (Corollary 8), and the sequence $\{\min(\mathbf{B}^t)\}$ is bounded above by 1 and non-decreasing. Both limits exist by the Monotone Convergence Theorem [18]. $\square$

3.2 Scaled Linear Bias Functions

Definition 7 (Scaled linear bias family). The *scaled linear bias family* is

$$\mathcal{F}_\alpha = \{f_\alpha : [-1, 1] \rightarrow [-1, 1] \mid f_\alpha(x) = \alpha x, \alpha \in (0, 1)\}.$$

The scalar $\alpha \in (0, 1)$ encodes the degree of receptiveness: as $\alpha \rightarrow 1$, f_α approaches the DeGroot identity; as $\alpha \rightarrow 0$, it approaches the insular function. Figure 1 shows the family alongside the DeGroot identity and the confirmation bias function of [17] on the square $[-1, 1]^2$.

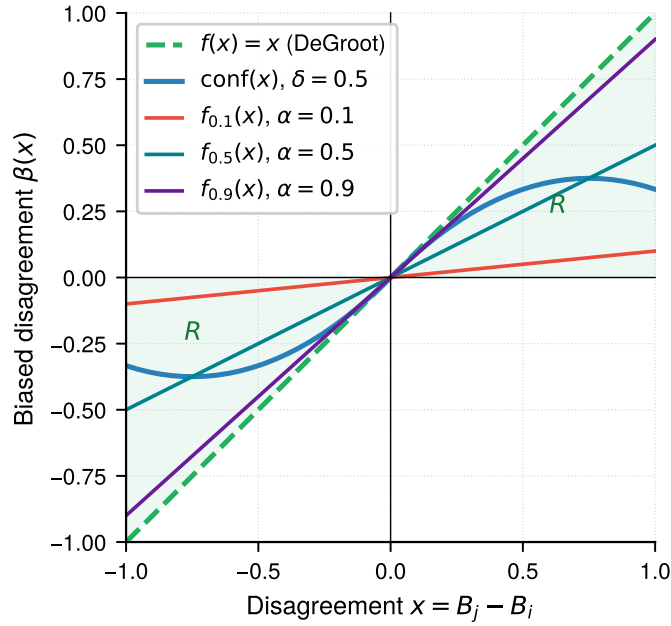


Figure 1: Bias functions on the square $[-1, 1]^2$. The identity $f(x) = x$ (DeGroot, green) marks the boundary between the Malleability region M and the Receptive-Resistant region R . The confirmation bias $\text{conf}(x)$ (blue, [17]) exhibits nonlinear resistance to distant opinions. Three members of $\mathcal{F}_\alpha$ are shown with $\alpha \in \{0.1, 0.5, 0.9\}$ (magenta, cyan, dark-red). In the first quadrant all scaled functions lie strictly below the identity, and in the third quadrant strictly above it, confirming membership in R .

Lemma 10 (Scaled linear biases lie in R). *For every $\alpha \in (0, 1)$, $f_\alpha \in R$.*

Proof. For $x > 0$: $f_\alpha(x) = \alpha x \in (0, x)$, so $(x, f_\alpha(x)) \in R$. For $x < 0$: $f_\alpha(x) = \alpha x \in (x, 0)$, so $(x, f_\alpha(x)) \in R$. For $x = 0$: $f_\alpha(0) = 0$, so $(0, 0) \in R$. $\square$

By Lemma 10, all results of Section 3.1 apply uniformly to $\mathcal{F}_\alpha$. Moreover, on cliques, $\mathcal{F}_\alpha$ -biases induce a *contraction* of the opinion spread $\Delta_t = \max(\mathbf{B}^t) - \min(\mathbf{B}^t)$.

Lemma 11 (Geometric decay of opinion spread). *Let $(G, \mathbf{B}^0, \mathbf{S}^0, \mu_G)$ be a CBSOM⁻ model on a clique with $n \geq 2$, with $\beta_{i,j} = f_{\alpha_{i,j}}$ for all edges. Then for every $\varepsilon > 0$ there exists $t \in \mathbb{N}$ such that $\Delta_t < \varepsilon$.*

Proof sketch. Let $\alpha_{\min} = \min_{(j,i) \in E} \alpha_{i,j}$ and $I_{\min} = \min_{(j,i) \in E} I_{ji}$. Consider a round in which a vocal neighbor j influences the max-agent i . All terms in the update of the max-agent are non-positive, so the maximum decreases by at least $I_{\min} \cdot \alpha_{\min} \cdot \Delta_t$. By Corollary 6, such rounds occur infinitely often. The full proof (Appendix A) also handles the case in which the min-agent is persistently silent: vocal agents then pull its opinion towards the maximum, which again forces $\Delta_t \rightarrow 0$. $\square$

3.3 Consensus in Clique Graphs

3.3.1 Cliques with $n \geq 3$

Theorem 12 (Consensus in CBSOM⁻ cliques with $n \geq 3$). *Let $(G, \mathbf{B}^0, \mathbf{S}^0, \mu_G)$ be a CBSOM⁻ model on an n -agent clique with $n \geq 3$ and $\beta_{i,j} = f_{\alpha_{i,j}} \in \mathcal{F}_\alpha$ for all edges. Then all agents converge to consensus.*

Proof sketch. By Theorem 9, limits U and L exist. Suppose $U - L > 0$. By Lemma 11, $\Delta_t < U - L$ for some t ; yet Corollary 8 requires $\Delta_t \geq U - L$ for all t , a contradiction. Hence $U = L$, and the Squeeze Theorem [18] yields consensus. The full proof is in Appendix A. $\square$

Remark 3.1. Theorem 12 extends Aranda et al.’s consensus result for SOM⁻ cliques [13] (identity bias, $\alpha = 1$) to the full scaled linear family $\mathcal{F}_\alpha$, and lifts Alvim et al.’s R -bias consensus result [17] from the synchronous DeGroot setting to the Spiral-of-Silence setting.

Figures 2 and 3 illustrate consensus convergence on a 3-agent clique under $\text{conf}(x)$ and f_α biases, respectively.

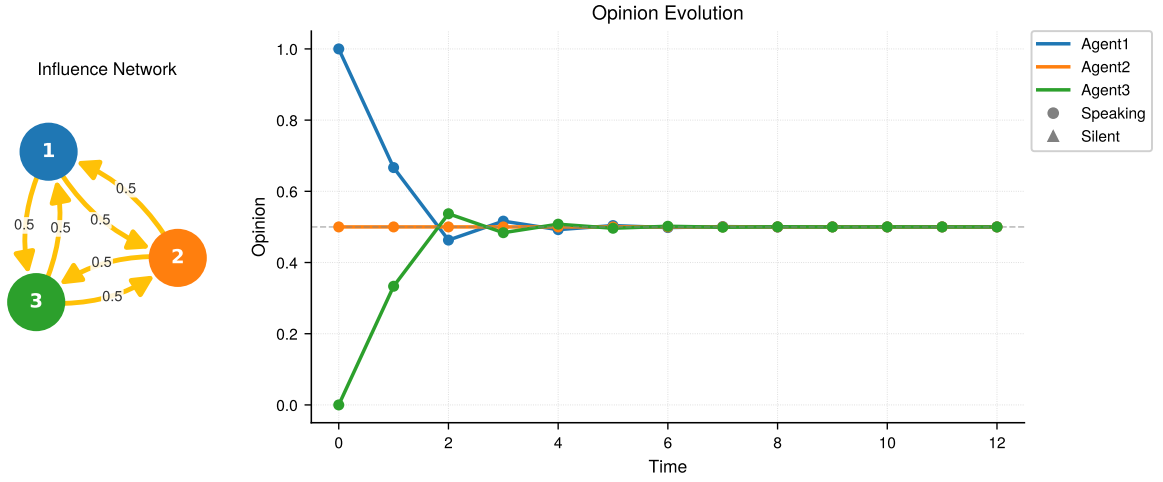


Figure 2: Consensus on a 3-agent clique under CBSOM⁻ with $\text{conf}(x)$ bias on all edges (Theorem 12). All directed edges carry influence weight 0.5; initial beliefs are $\mathbf{B}^0 = (1, 0.5, 0)$; $\tau_i = 1$

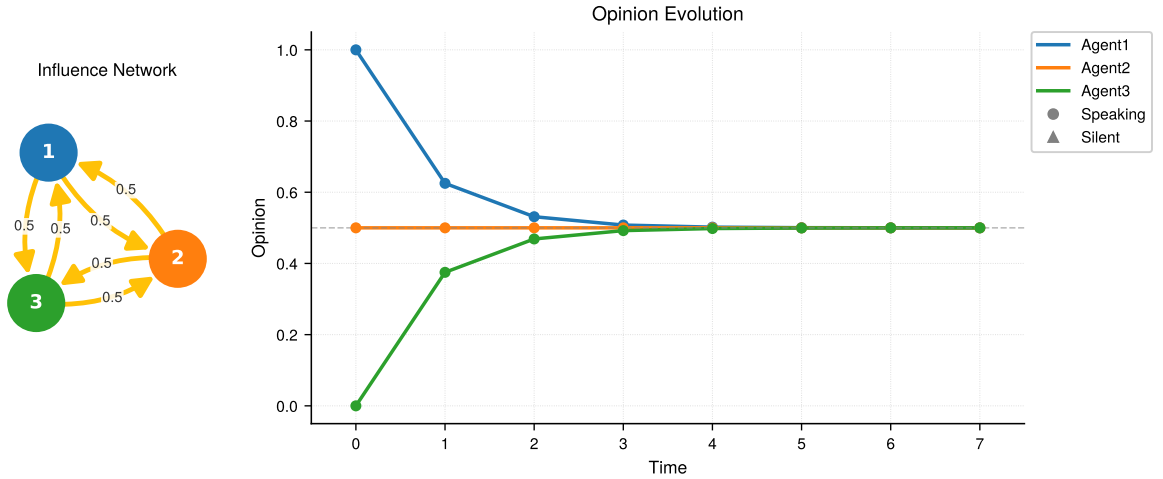


Figure 3: Consensus on a 3-agent clique under CBSOM⁻ with scaled linear bias f_α , $\alpha = 0.5$ (Theorem 12). Same parameters as Figure 2. Consensus is reached faster than under $\text{conf}(x)$, because f_α attenuates disagreements of every size linearly

3.3.2 The Dyadic Case $n = 2$

On two-agent cliques, the model SOM⁻ can fail to converge and exhibit period-2 oscillations [13]. Figure 4 illustrates this pathology. With symmetric influence weights $I_{12} = I_{21} = 1.0$ and $\tau_i = 1$, the agents alternate

their beliefs between 0 and 1 forever and never reach agreement.

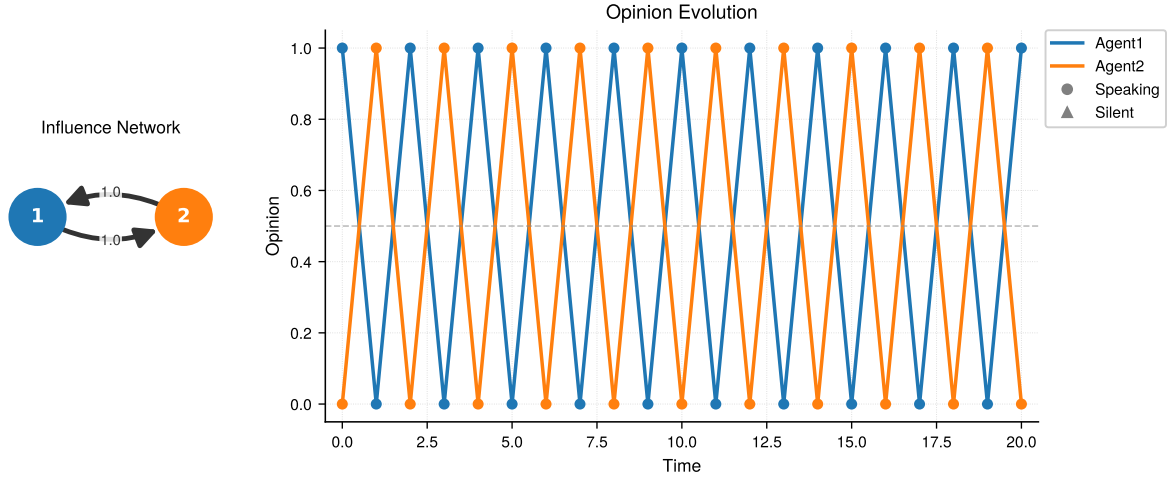


Figure 4: Non-convergence counterexample for SOM^- on a dyadic clique ($n = 2$). Symmetric influence weights $I_{12} = I_{21} = 1.0$, $\mathbf{B}^0 = (1, 0)$, $\tau_i = 1$. Agents permanently oscillate between 0 and 1, alternating speaking and silent roles; consensus is never reached. This is the pathology that Theorem 14 resolves by introducing $\mathcal{F}_\alpha$ -biases

We now show that $\mathcal{F}_\alpha$ -biases eliminate this pathology.

Lemma 13 (Persistent vocality for $n = 2$). *Consider CBSOM^- on two agents $\{1, 2\}$ with $\tau_1 = \tau_2 = 1$. Then $S_1^t = S_2^t = 1$ for all $t \in \mathbb{N}$.*

Proof. With $\tau_i = 1$, the condition $|B_i^t - B_j^t| \leq 1$ holds for all t since opinions lie in $[0, 1]$. Hence every agent always satisfies the majority criterion and remains vocal. $\square$

Theorem 14 (Consensus under scaled linear biases for $n = 2$). *Consider CBSOM^- on a two-agent clique $\{1, 2\}$ with $I_{12}, I_{21} > 0$, biases $\beta_{i,j}(x) = \alpha_{i,j}x$ with $\alpha_{i,j} \in (0, 1)$, and $\tau_1 = \tau_2 = 1$. Then both agents converge to consensus.*

Proof. By Lemma 13, $S_j^t = 1$ for all t . Let $\lambda_{12} = I_{21}\alpha_{1,2}$ and $\lambda_{21} = I_{12}\alpha_{2,1}$. Eq. (8) reduces to:

$$\begin{aligned} B_1^{t+1} &= (1 - \lambda_{12})B_1^t + \lambda_{12}B_2^t, \\ B_2^{t+1} &= \lambda_{21}B_1^t + (1 - \lambda_{21})B_2^t. \end{aligned}$$

Let $D^t = B_1^t - B_2^t$. Subtracting: $D^{t+1} = \gamma D^t$ where $\gamma = 1 - \lambda_{12} - \lambda_{21}$. Since $\alpha_{i,j} \in (0, 1)$ and $I_{ji} > 0$ for both edges, $\lambda_{12} = I_{21}\alpha_{1,2} \in (0, I_{21})$ and $\lambda_{21} = I_{12}\alpha_{2,1} \in (0, I_{12})$ are both strictly positive. Hence $\lambda_{12} + \lambda_{21} \in (0, I_{21} + I_{12})$. Since $I_{12}, I_{21} \leq 1$ and $\alpha_{i,j} < 1$, $\lambda_{12} + \lambda_{21} < 2$; combined with strict positivity, $\gamma = 1 - \lambda_{12} - \lambda_{21} \in (-1, 1)$, so $|\gamma| < 1$. Therefore $D^t = \gamma^t D^0 \rightarrow 0$, and both agents converge to the same limit $v \in [0, 1]$. $\square$

Remark 3.2 (Asymmetric bias case). Theorem 14 requires $\alpha_{i,j} \in (0, 1)$ for *both* directed edges. The asymmetric case—one edge with $\alpha = 1$ (identity, no bias) and one with $\alpha \in (0, 1)$ —falls outside the stated hypotheses but is conjectured to converge; we leave a formal treatment for future work.

Remark 3.3 (Recovery of the $n = 2$ pathology). The period-2 oscillation of SOM^- [13] corresponds to $\gamma = -1$, requiring $\lambda_{12} + \lambda_{21} = 2$. With $\alpha_{i,j} \in (0, 1)$, $\lambda_{ij} = I_{ji}\alpha_{i,j} < 1$ strictly, so $|\gamma| < 1$ always. In the symmetric case ($I_{12} = I_{21} = I$, $\alpha_{1,2} = \alpha_{2,1} = \alpha$), the contraction rate is $\gamma = 1 - 2\alpha I$. If $2\alpha I > 1$, then $\gamma < 0$ and the dynamics oscillate but still converge; if $2\alpha I < 1$, then $\gamma > 0$ and the convergence is monotone.

Figures 5 and 6 illustrate how α controls the qualitative character of the convergence guaranteed by Theorem 14.

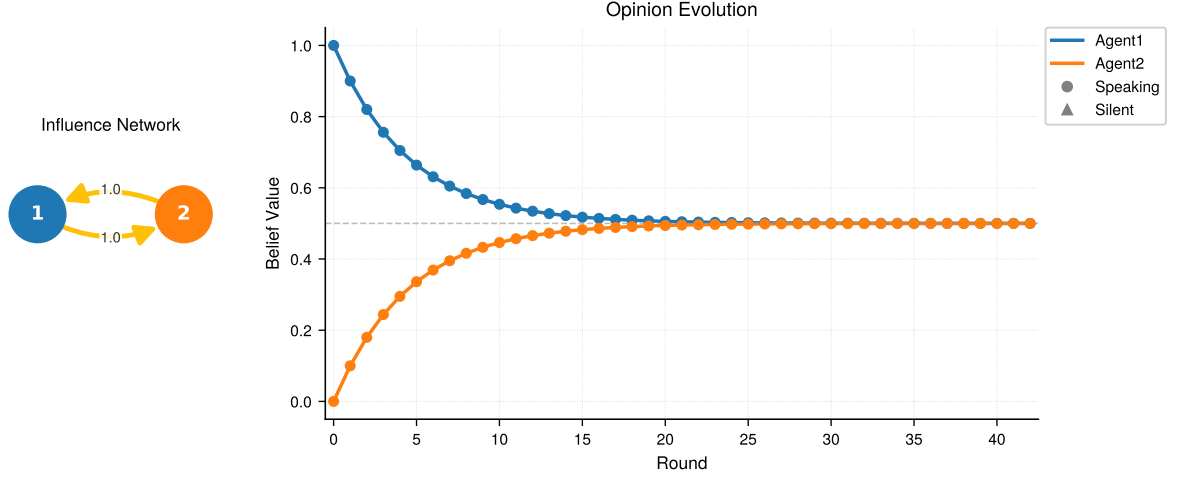


Figure 5: Consensus on a two-agent clique under CBSOM^- with scaled linear bias f_α , $\alpha = 0.1$ (Theorem 14). Parameters: $I_{12} = I_{21} = 1.0$, $\mathbf{B}^0 = (1, 0)$, $\tau_i = 1$. With $\gamma = 1 - 2\alpha I = 0.8$, the opinion gap shrinks monotonically (“inverse trumpet” profile)

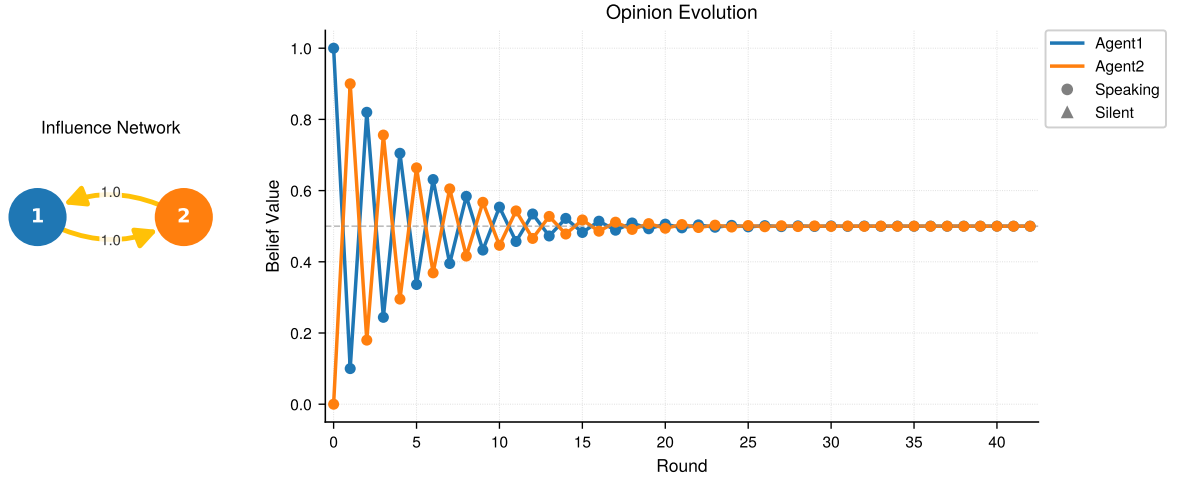


Figure 6: Consensus on a two-agent clique under CBSOM^- with scaled linear bias f_α , $\alpha = 0.9$. Same parameters as Figure 5. With $\gamma = -0.8$, the gap oscillates with strictly decreasing amplitude. The behavior resembles the SOM^- pathology (Figure 4), but it converges. The special case $\alpha = 0.5$ reaches consensus exactly at round 1: $B_1^1 = B_2^1 = 0.5$

3.4 Summary

Table 2 summarizes the consensus landscape across models and clique sizes.

Table 2: Consensus in clique graphs: SOM^- vs. CBSOM^- with $\mathcal{F}_\alpha$ -biases

Model	$n = 2$	$n \geq 3$
SOM^- [13]	oscillates	converges
$\text{CBSOM}^- + \mathcal{F}_\alpha$ (<i>this work</i>)	converges	converges

The decoupling between bias and silence, formalized in Lemma 5, is the structural key: silence determines *who speaks*, while bias determines *how the spoken opinion is absorbed*. Consensus emerges from the joint action of the contraction induced by $\mathcal{F}_\alpha$ (Lemma 11) and the impossibility of perpetual silence (Corollary 6).

4 Experimental Evaluation

The results of Section 3 cover CBSOM^- on cliques with scaled linear biases. In this section, we study through simulation the settings that those results do not cover: scale-free graphs, biases outside the region R , mixtures

of biases, and the memory-based model CBSOM⁺. We run both models across eight network sizes and a full range of connectivity levels. Our goals are threefold: (i) to test whether only R -region bias functions support consensus; (ii) to quantify the joint effect of network size and density on the consensus threshold; and (iii) to characterize how memory changes these effects.

4.1 Simulation Setup

The simulator generates scale-free graphs by preferential attachment with degree-distribution exponent $\gamma = 2.5$. This value lies in the empirically documented range for scale-free social networks [19–21]. The parameter grid is:

1. **Network sizes:** $n \in \{10, 50, 100, 500, 10^3, 10^4, 10^5, 10^6\}$.
2. **Density:** 1–9 average neighbors per agent.
3. **Bias types:** Unbiased (SOM[−]/SOM⁺), Confirmation, Backfire, and Authority (all under CBSOM[−] or CBSOM⁺).
4. **Runs per configuration:** 1,000 ($n \leq 10^3$); 500 ($n = 10^4$); 50 ($n = 10^5$); 10 ($n = 10^6$).
5. **Convergence criterion:** opinion spread $\max_i B_i^t - \min_i B_i^t < 10^{-3}$; iteration cap 1,000 rounds.
6. **Initial opinions:** $B_i^0 \sim \mathcal{U}[0, 1]$.

Metrics. The *Consensus Rate* CR is the fraction of runs that converge. The *Mean Convergence Time* MCT is the average number of rounds over converging runs. We report CR throughout.¹

4.2 Uniform Networks

A *uniform* network assigns the same bias type to every edge and the same silence model to every agent. This setting isolates the contribution of each bias region before we consider mixtures.

Memoryless dynamics (CBSOM[−]). Figure 7 (top row) summarizes the full $(n, \text{density})$ grid. Both SOM[−] and CBSOM[−] (Confirmation) move from near-zero to near-100% consensus as density increases. The Confirmation panel is always at least as light as the SOM[−] panel: R -region biases act as *consensus accelerators*. The advantage is largest at the transition. At $n = 10^6$ and density 4, the consensus rate is 70% for CBSOM[−] (Confirmation) and 10% for SOM[−] (Figure 8). The Backfire panel is uniformly black, i.e., there is no consensus at any scale or density. Authority also yields zero consensus, except for negligible events in a 10-agent network. This pattern is consistent with Theorems 12 and 14, which guarantee consensus only for R -region biases. However, at $n = 10^6$ the sample is too small (10 runs per configuration) to assert an absolute dichotomy; these results should be taken as indicative.

The consensus threshold grows sub-linearly with n . Density 2 suffices for $n \leq 100$, density 3–4 for $n \in [10^3, 10^5]$, and density 4–5 for $n = 10^6$. Figure 8 shows the corresponding rightward shift of the S-curve: as n increases, the transition from zero to full consensus requires one or two additional neighbors per agent. Once the threshold is crossed, the consensus rate approaches 100%. This holds in all observed runs for $n \leq 10^5$, where each configuration has at least 50 runs.

Memory-based dynamics (CBSOM⁺). The bottom row of Figure 7 differs markedly from the top row. For $n \leq 100$, the Confirmation panel retains a faint non-zero region at low densities (e.g., 19.9% at $n = 50$, density 2), while SOM⁺ is already near zero. For $n = 500$, CBSOM⁺ (Confirmation) peaks at 4.9%; for $n = 1,000$, it falls to 3.8% and is confined to densities 4–7. Beyond $n = 10^4$, no memory-based configuration, including Confirmation, reaches consensus in a single run at any density. Figure 9 makes this collapse visible. The Confirmation advantage that sustains high consensus in the left panel (CBSOM[−]) disappears in the right panel (CBSOM⁺) as n grows.

¹All experiments were run on an Intel i7-1255U machine with 32 GB RAM and a 512 GB NVMe SSD.

Consensus rate — uniform networks (density $1-9 \times$ network size)

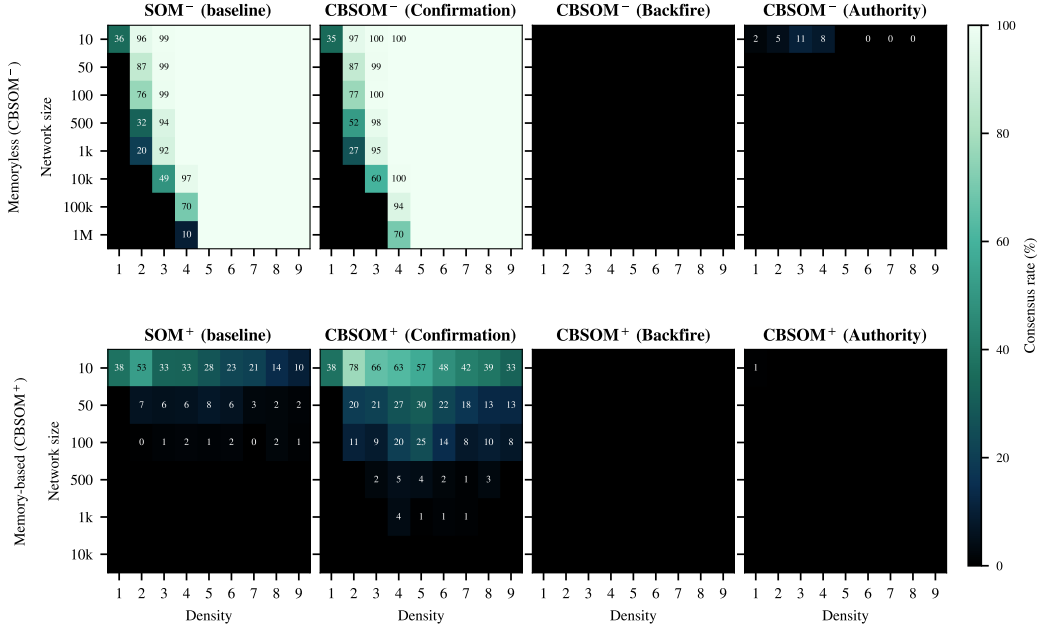


Figure 7: Consensus rate (color, black = 0%, near-white = 100%) for *uniform* networks over the full $(n, \text{density})$ grid. **Top row** (CBSOM[−], memoryless): SOM[−] and Confirmation transition to near-white at moderate density across all eight scales; Backfire and Authority are uniformly black. **Bottom row** (CBSOM⁺, memory-based): only Confirmation retains a faint non-zero patch for $n \leq 100$ and low density; all other configurations yield zero

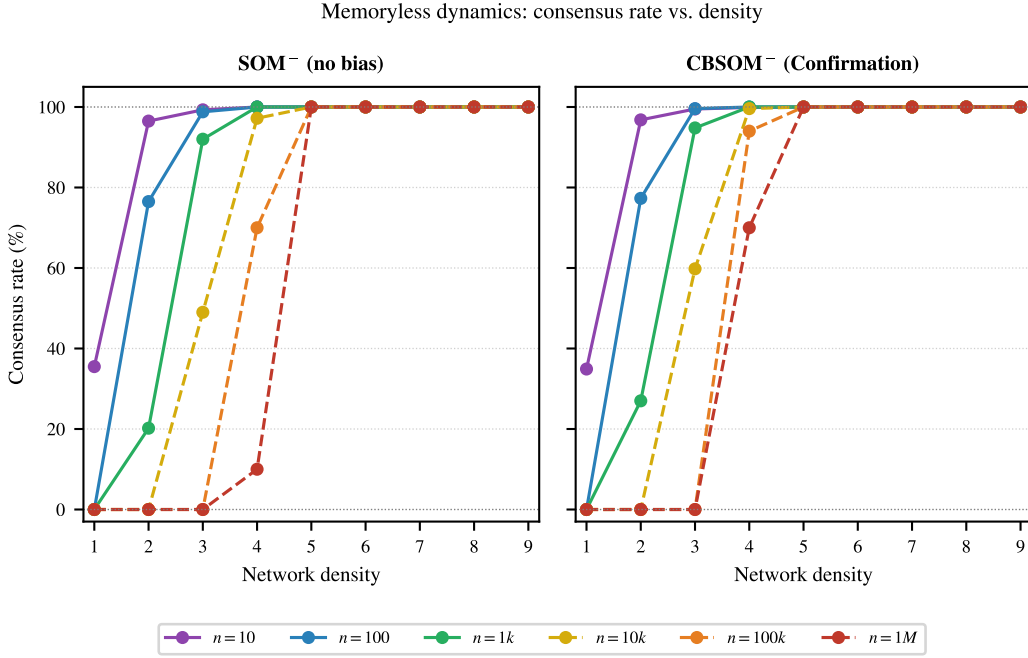


Figure 8: Consensus rate vs. density for SOM[−] (left) and CBSOM[−] (Confirmation, right) at six representative scales. Confirmation (right) saturates one density unit earlier than SOM[−] (left) for the same n , and the S-curve shifts rightward as n grows, reflecting the sub-linear increase in the consensus threshold with network size

Memory penalty: Confirmation bias under memoryless vs. memory-based dynamics

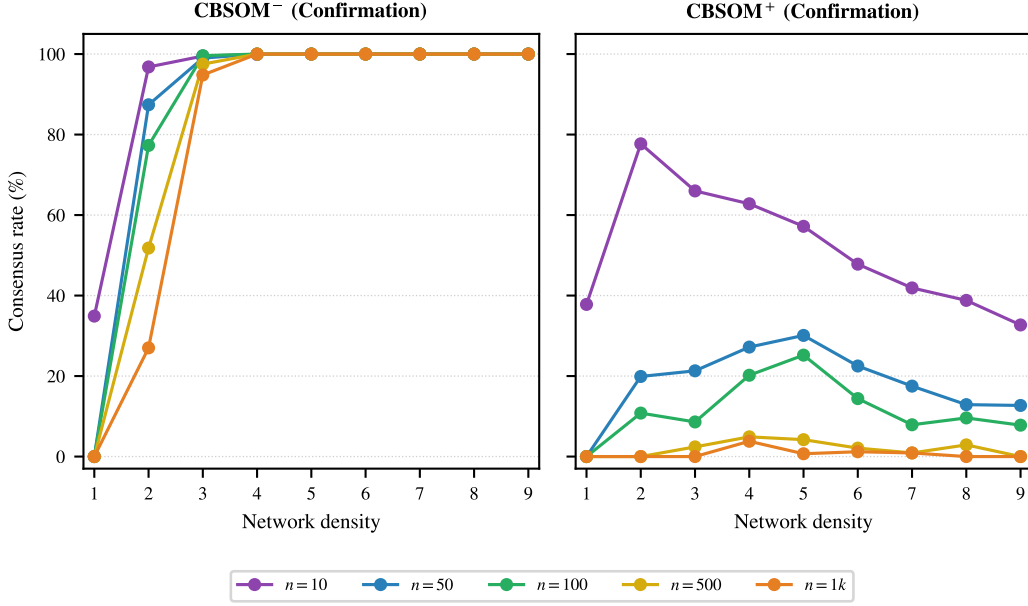


Figure 9: Memory penalty under Confirmation bias: CBSOM⁻ (left) versus CBSOM⁺ (right). In the memoryless model consensus is robust across all $n \leq 10^3$; in the memory-based model it collapses rapidly as n grows, vanishing entirely for $n = 1,000$ at high densities

4.3 Non-Uniform Networks

Non-uniform networks assign different bias types to different edges. We consider four mixed configurations; the proportions refer to fractions of edges:

- (i) SOM⁻ + CBSOM⁻ (Conf.), 70%/30%
- (ii) CBSOM⁻ (Conf.) + CBSOM⁻ (Back.), 80%/20%
- (iii) CBSOM⁻ (Conf.) + CBSOM⁻ (Auth.), 60%/40%
- (iv) SOM⁻ + CBSOM⁻ (Back.), 90%/10%

Rationale for the choice of proportions. We selected the four configurations heuristically. They are not calibrated to measured distributions of cognitive biases in real social networks; to our knowledge, no such data exist at the resolution required to assign a bias type to each edge. Instead, the proportions follow from the uniform-network results of Section 4.2 and from an observation already present in the Spiral-of-Silence model of Aranda et al. [13]: consensus is structurally difficult to achieve even without cognitive biases.

Uniform Backfire and Authority configurations yield zero consensus at every scale and density. Pure non- R configurations therefore offer no interesting boundary to explore. The relevant question is how small a disruptive minority suffices to destroy consensus that would otherwise form. Configuration (iv) (10% Backfire) is a *contamination* scenario: it asks whether a marginal fraction of adversarial edges can override an otherwise consensus-forming majority. Configuration (ii) (20% Backfire) raises the disruptive load but keeps a receptive-resistant supermajority. It probes the point at which updating dominated by $\text{conf}(x)$ can no longer absorb the opposing force. Configuration (iii) (40% Authority) places the two regimes in near-balance. It tests whether substantial hierarchical influence is compatible with global agreement. Finally, configuration (i) (30% $\text{conf}(x)$, 70% unbiased) asks the complementary question: can a receptive-resistant minority *rescue* or amplify consensus in an unbiased network that already struggles to converge at large scales?

The proportions thus span a gradient from marginal perturbation to near-parity. They cover regimes in which consensus is expected to be fragile, recoverable, or structurally precluded. Bias distributions that vary by agent role, community membership, or time may shift these thresholds. We leave the calibration of proportions to empirical social-network data for future work.

Memoryless dynamics. Figure 10 shows the results. Configuration (i) closely mirrors the uniform Confirmation baseline. Consensus reaches 97% by density 3 for $n = 500$, and the transition shifts rightward

to density 5 only for $n = 10^6$. Hence mixing confirmation-biased and unbiased edges in a 30/70 proportion preserves the robustness of consensus.

Configurations (ii)–(iv) collapse to near-zero consensus at every scale. At $n = 100$, configurations (ii) and (iv) produce at most 2%–3% consensus across all densities; for $n \geq 500$, consensus is identically zero. A 10% share of Backfire edges (configuration iv) suffices to eliminate consensus that would otherwise occur in almost every run. This *robustness asymmetry* has a clear direction. Edges in R act as consensus catalysts, whereas edges outside R prevent consensus, and a surrounding majority of R -region edges cannot neutralize them.

Memory-based dynamics. In non-uniform memory-based networks, consensus is essentially absent. The $\text{SOM}^+ + \text{CBSOM}^+(\text{Conf.})$ 70/30 mixture produces sporadic consensus only for $n \leq 100$ (at most 30%, density 4). For $n \geq 500$, no run converges under any mixed memory-based configuration. Memory thus amplifies the disruptive effect of non- R edges: at scale, we observed no consensus at all.

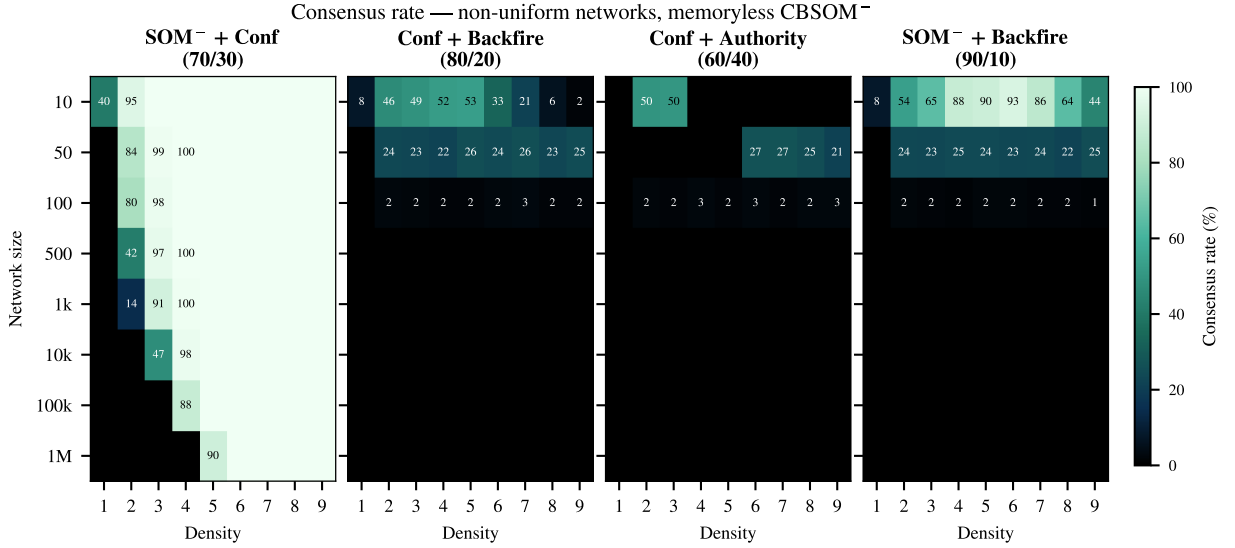


Figure 10: Consensus rate for *non-uniform* memoryless networks (CBSOM^-) over the full $(n, \text{density})$ grid. Only the $\text{SOM}^- + \text{CBSOM}^- (\text{Conf.})$ 70/30 mixture (leftmost panel) achieves near-100% consensus at moderate density. Every configuration that contains Backfire or Authority edges collapses to near-zero consensus at every scale. This illustrates the robustness asymmetry between influence inside and outside the region R .

4.4 Relative Density

The experiments in Sections 4.2 and 4.3 measure connectivity by an absolute neighbor count $k \in \{1, \dots, 9\}$. The same count represents very different levels of social exposure as the network grows. To assess whether the observed effects depend on this choice, we add a *relative-density* analysis. In it, the neighborhood of each agent is a fixed fraction $\rho \in \{10\%, 25\%, 50\%, 75\%, 100\%\}$ of the maximum feasible degree $n - 1$. The 100% condition is the complete-graph limit: every agent interacts with every other. Experiments cover $n \in \{100, 1,000, 10,000\}$ with the same run counts and convergence criterion as Section 4.1.

Figures 11 and 12 summarize the four scenarios, one per figure row.

Uniform memoryless (Figure 11, top row). SOM^- and CBSOM^- (Confirmation) achieve 100% consensus at every relative density and every tested scale, including the sparsest configuration ($\rho = 10\%$, $n = 10,000$, i.e. $\approx 1,000$ neighbors per agent). Backfire and Authority yield 0% everywhere. The all-white and all-black panels indicate that, in this range, the bias-region dichotomy depends on the bias type alone and not on the connectivity level.

Uniform memory-based (Figure 11, bottom row). All cells are black for $n \geq 1,000$ across all configurations and all relative densities, including the complete-graph limit ($\rho = 100\%$). The only non-zero values appear at $n = 100$: 11.1% and 0.9% for Conf + Backfire (80/20) at $\rho = 10\%$ and 25% respectively, and 0.1% for $\text{SOM}^+ + \text{Confirmation}$ (70/30) at $\rho = 10\%$; all vanish at $\rho \geq 50\%$.

Non-uniform memoryless (Figure 12, top row). The $\text{SOM}^- + \text{Conf}$ (70/30) column is entirely white: mixing a 30% confirmation minority with an unbiased majority preserves 100% consensus at every scale and every relative density. The three disruptive mixtures annotate small non-zero values (1.4–3.0%) only at $n = 100$; for $n \geq 1,000$ they are identically zero regardless of ρ .

Non-uniform memory-based (Figure 12, bottom row). The entire row is black: zero consensus for all configurations, all three scales, and all relative densities up to the complete graph.

Together, the two memory-based scenarios sharpen the memory-barrier conclusion of Section 4.5. Increasing connectivity, up to and including the complete graph, does not overcome the suppression of consensus under CBSOM^+ . The barrier is therefore structural rather than a threshold effect.

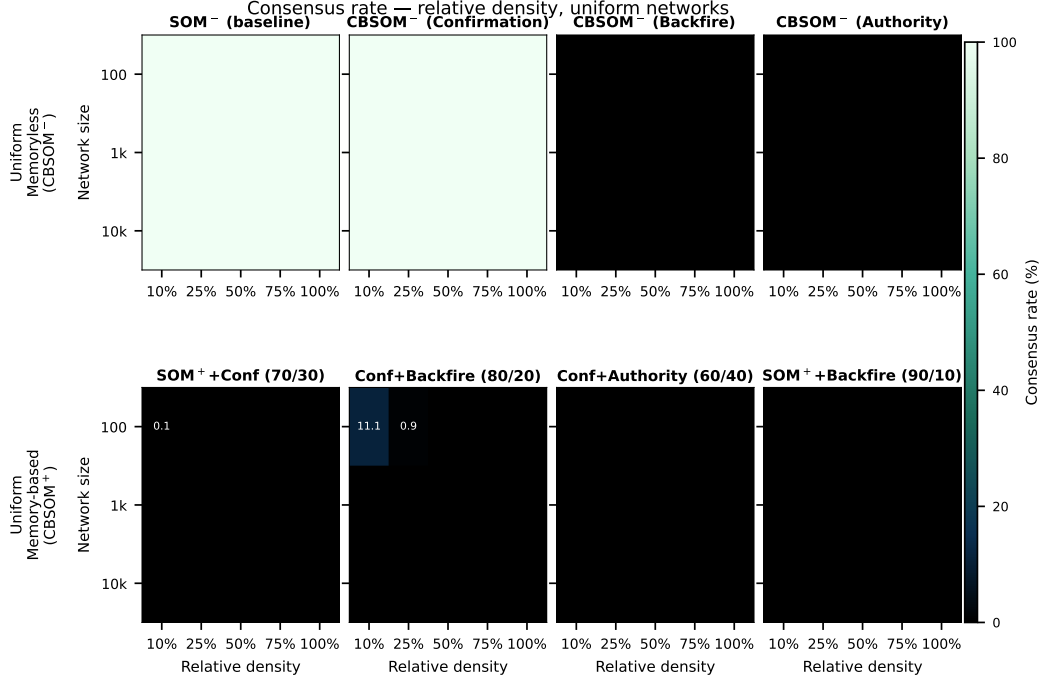


Figure 11: Consensus rate (color scale as in Figures 7–10) for relative-density experiments on *uniform* networks ($\rho \in \{10\%, 25\%, 50\%, 75\%, 100\%\}$, $n \in \{100, 1k, 10k\}$). **Top row** (memoryless): SOM^- and Confirmation are near-white at every cell; Backfire and Authority are uniformly black; the bias dichotomy does not depend on density. **Bottom row** (memory-based): all black for $n \geq 1,000$ at every ρ , including the complete graph ($\rho = 100\%$); non-zero values appear only at $n = 100$, $\rho \leq 25\%$ (11.1% and 0.9% for Conf+Backfire; 0.1% for $\text{SOM}^+ + \text{Conf}$)

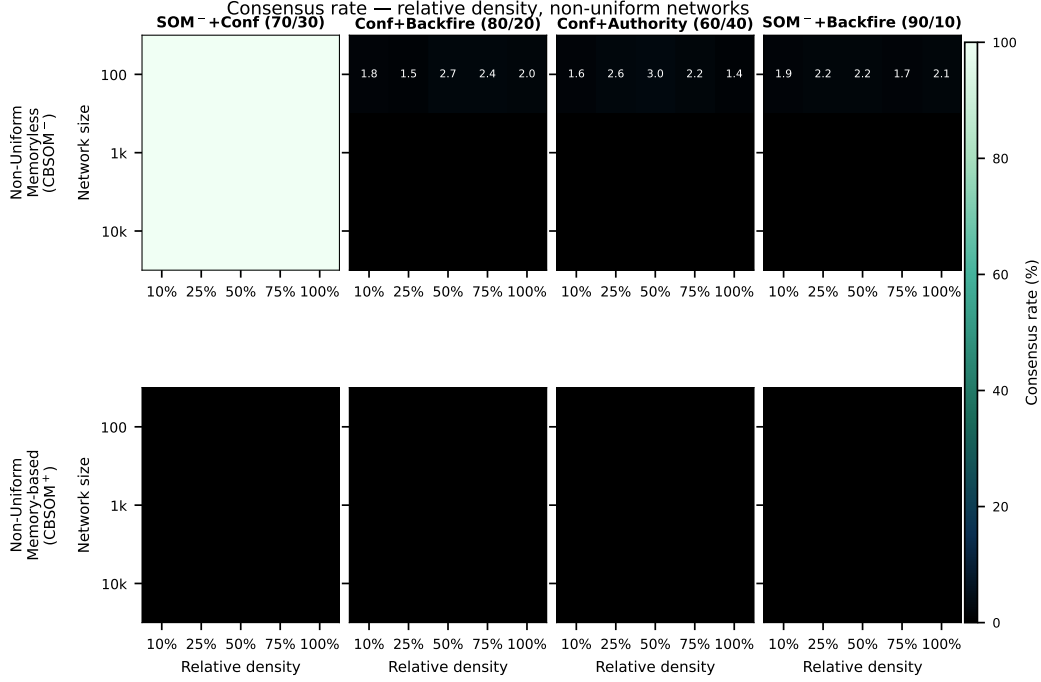


Figure 12: Consensus rate for relative-density experiments on *non-uniform* networks (same color scale and parameter grid as Figure 11). **Top row** (memoryless): the $\text{SOM}^- + \text{Conf}$ (70/30) column is uniformly near-white at all scales and relative densities; disruptive mixtures annotate marginal values ($\leq 3\%$) only at $n = 100$, vanishing for $n \geq 1,000$. **Bottom row** (memory-based): uniformly black across all configurations, scales, and relative densities, including the complete-graph limit ($\rho = 100\%$)

4.5 Discussion

Three empirical findings summarize the experimental evidence:

1. **Bias-region dichotomy.** R -region biases (Confirmation) yield CR approaching 100% at sufficient density for $n \leq 10^5$ (at least 50 runs per configuration). At $n = 10^6$ (10 runs), all Confirmation runs converge at density ≥ 5 , but the sample is too small to assert $\text{CR} = 100\%$ unconditionally. Non- R biases (Backfire, Authority) yield $\text{CR} = 0\%$ in all observed runs at every scale and density tested.
2. **Sub-linear consensus threshold.** The critical density grows from 2 (small networks) to 4–5 ($n = 10^6$), spanning only 3 units across eight orders of magnitude of network size. The threshold under Confirmation is consistently one unit lower than under the unbiased SOM^- baseline.
3. **Memory as a structural barrier.** Memory-based dynamics (SOM^+ , CBSOM^+) almost entirely suppress consensus. Even Confirmation retains non-zero CR only at $n \leq 1,000$. In non-uniform settings, an R -region majority cannot compensate for the path dependence induced by memory beyond $n = 100$.

Data and code availability. The Scala/Akka simulator used to produce the results reported in this section is publicly available at https://github.com/juanmarcosdev/belief_evolution_simulator.

5 Conclusions and Related Work

We introduced CBSOM^- and CBSOM^+ , two extensions of the Spiral-of-Silence opinion models of [13] that incorporate cognitive bias functions into the belief-update rule. We showed that, in the memoryless model, the Receptive-Resistant region R preserves the structural properties of the base model: global silence never lasts more than one round, and the extreme opinions contract monotonically. For scaled linear biases, we proved consensus on cliques of every size $n \geq 2$. Large-scale simulations, with networks from 10 to 10^6 agents, complement these results. They show a dichotomy between bias regions, a sub-linear growth of the critical density with network size, and a consensus barrier introduced by memory. To our knowledge, this is the first work that embeds cognitive bias functions into a Spiral-of-Silence framework and gives formal consensus guarantees for the resulting models.

The work closest to ours is that of Alvim et al. [17]. It introduces the taxonomy of bias regions used throughout this paper (receptive-resistant, malleable, backfire, and insular), and proves that continuous

R -region biases guarantee consensus in the synchronous DeGroot model on strongly connected graphs. Our Theorem 12 lifts this guarantee to the Spiral-of-Silence setting, where the silence mechanism decouples from the bias layer. Unlike [17], our consensus result is restricted to cliques and to the scaled linear family $\mathcal{F}_\alpha$. Extending it to all continuous R -biases on arbitrary strongly connected graphs remains an open problem. The psychological foundations of the confirmation, backfire, and authority biases considered here are discussed in [14].

The models SOM^- and SOM^+ were introduced in [13]. That work establishes consensus in n -agent cliques for $n \geq 3$ and exhibits the period-2 oscillation on two-agent cliques. Our Theorem 14 shows that every scaled linear bias f_α with $\alpha \in (0, 1)$ breaks this oscillation and restores consensus. In our simulations, the memory-based model CBSOM^+ reaches consensus far less often than CBSOM^- at every scale. This mirrors the behavior observed for SOM^+ in [13], and suggests that memory acts as a barrier to consensus independently of the bias layer. The decoupling identified in Section 3.1 explains why the guarantees of [13] carry over to CBSOM^- : the silence criterion depends only on raw opinion proximity, so the bias layer does not interfere with the conditions that produce consensus in the base model. A comparable argument for CBSOM^+ remains open, because memory couples historical opinion values with the current silence decision.

Bounded-confidence models [22] let agents interact only with those whose opinions lie within a confidence threshold. The tolerance radius τ_i of our models plays a similar role, but it governs whether an agent speaks, not whom it listens to. Agent-based simulations have also been used to study the Spiral of Silence in online settings [9–12]. Our experimental study follows this line, and complements it with formal results for the clique case.

In future work, we plan to extend cognitive biases to the confidence-based silence models of Gaona [23], which have not yet been studied under cognitive bias. We also plan to study dynamic network topologies, in which the influence graph evolves over time.

Acknowledgment

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A Proofs

This appendix contains the complete proofs of the following results:

- **Lemma 7.** (Bounded opinions under CBSOM[−])
- **Lemma 11.** (Geometric decay of opinion spread)
- **Theorem 12.** (Consensus in CBSOM[−] cliques with $n \geq 3$)

Lemma 7 (Bounded opinions under CBSOM[−]). *Let $(G, \mathbf{B}^0, \mathbf{S}^0, \mu_G)$ be a CBSOM[−] model with continuous bias functions $\beta_{i,j} \in \mathbb{R}$ for all edges. For every $t \in \mathbb{N}$ and all $i \in A$:*

$$\min(\mathbf{B}^t) \leq B_i^{t+1} \leq \max(\mathbf{B}^t).$$

Proof. Let $M = \max(\mathbf{B}^t)$ and $m = \min(\mathbf{B}^t)$. Fix an agent $i \in A$. By Eq. (8):

$$B_i^{t+1} = B_i^t + \underbrace{\sum_{j \in N^t(i)} I_{ji} \cdot \beta_{i,j}(B_j^t - B_i^t)}_{=: \Sigma_i^t}, \quad (12)$$

where $N^t(i) = \{j \in N(i) : S_j^t = 1\}$. We partition $N^t(i)$ into three disjoint sets:

$$\begin{aligned} P_i^+ &= \{j \in N^t(i) : B_j^t > B_i^t\}, \\ P_i^- &= \{j \in N^t(i) : B_j^t < B_i^t\}, \\ P_i^0 &= \{j \in N^t(i) : B_j^t = B_i^t\}. \end{aligned}$$

For every $j \in P_i^0$ the term is zero since $\beta_{i,j}(0) = 0$ (as $(0, 0) \in R$). We analyze the non-zero contributions.

Claim 1 (Sign and magnitude of $\beta_{i,j}$ in R). *For $\beta_{i,j} \in R$ and $x \neq 0$: (a) $\text{sign}(\beta_{i,j}(x)) = \text{sign}(x)$; (b) $|\beta_{i,j}(x)| \leq |x|$.*

Proof of Claim. Property (a) follows from the definition of R (the graph of $\beta_{i,j}$ lies in the first and third quadrants of the square $[-1, 1]^2$ [17]). Property (b) follows from $|\beta_{i,j}(x)| \leq |x|$ for all $(x, y) \in R$ with $y = \beta_{i,j}(x)$ [17]. $\square$

Upper bound: $B_i^{t+1} \leq M$.

By Claim 1(a): each term with $j \in P_i^-$ satisfies $I_{ji} \cdot \beta_{i,j}(B_j^t - B_i^t) \leq 0$. Dropping these non-positive contributions:

$$\Sigma_i^t \leq \sum_{j \in P_i^+} I_{ji} \cdot \beta_{i,j}(B_j^t - B_i^t). \quad (13)$$

By Claim 1(b), for $j \in P_i^+$: $\beta_{i,j}(B_j^t - B_i^t) \leq B_j^t - B_i^t$. Hence:

$$\begin{aligned} B_i^{t+1} &\leq B_i^t + \sum_{j \in P_i^+} I_{ji}(B_j^t - B_i^t) \\ &= B_i^t \left(1 - \sum_{j \in P_i^+} I_{ji}\right) + \sum_{j \in P_i^+} I_{ji} B_j^t. \end{aligned} \quad (14)$$

By row-stochasticity (Eq. (1)), $\sum_{j \in P_i^+} I_{ji} \leq \sum_{j \in N(i) \cup \{i\}} I_{ji} = 1$, so the coefficient of B_i^t is non-negative. Expression (14) is therefore a convex combination of B_i^t and $\{B_j^t\}_{j \in P_i^+}$, all of which lie in $(-\infty, M]$. Therefore $B_i^{t+1} \leq M$.

Lower bound: $B_i^{t+1} \geq m$.

By Claim 1(a): each term with $j \in P_i^+$ satisfies $I_{ji} \cdot \beta_{i,j}(B_j^t - B_i^t) \geq 0$. Dropping these non-negative contributions:

$$\Sigma_i^t \geq \sum_{j \in P_i^-} I_{ji} \cdot \beta_{i,j}(B_j^t - B_i^t). \quad (15)$$

By Claim 1(b), for $j \in P_i^-$ (where $B_j^t - B_i^t < 0$): $\beta_{i,j}(B_j^t - B_i^t) \geq B_j^t - B_i^t$. Hence:

$$\begin{aligned} B_i^{t+1} &\geq B_i^t + \sum_{j \in P_i^-} I_{ji}(B_j^t - B_i^t) \\ &= B_i^t \left(1 - \sum_{j \in P_i^-} I_{ji}\right) + \sum_{j \in P_i^-} I_{ji} B_j^t. \end{aligned} \quad (16)$$

By the same row-stochasticity argument, (16) is a convex combination of B_i^t and $\{B_j^t\}_{j \in P_i^-}$, all of which lie in $[m, +\infty)$. Therefore $B_i^{t+1} \geq m$.

Combining both bounds: $\min(\mathbf{B}^t) \leq B_i^{t+1} \leq \max(\mathbf{B}^t)$ for all $i \in A$. $\square$

Lemma 11 (Geometric decay of opinion spread). *Let $(G, \mathbf{B}^0, \mathbf{S}^0, \mu_G)$ be a CBSOM⁻ model on a clique with $n \geq 2$, with $\beta_{i,j} = f_{\alpha_{i,j}}$ for all edges. Then for every $\varepsilon > 0$ there exists $t \in \mathbb{N}$ such that $\Delta_t < \varepsilon$.*

Proof. Let $c = I_{\min} \cdot \alpha_{\min} \in (0, 1)$, where $I_{\min} = \min_{(j,i) \in E} I_{ji}$ and $\alpha_{\min} = \min_{(j,i) \in E} \alpha_{i,j}$. Both are strictly positive by assumption. Let $\Delta_t = \max(\mathbf{B}^t) - \min(\mathbf{B}^t)$. By Corollary 8, $\{\Delta_t\}$ is non-increasing.

We identify two disjoint types of steps and show that the second type occurs infinitely often, each time contracting Δ_t by a factor of at least $(1 - c)$.

Claim 2 (Max-agent update on a clique with $\mathcal{F}_\alpha$ -bias). *Let G be a clique, $\beta_{i,j} = f_{\alpha_{i,j}}$, and $\Delta_t > 0$. If agent i achieves $\max(\mathbf{B}^t)$ and there exists a vocal neighbor $j \in N^t(i)$ with $B_j^t < B_i^t$, then:*

$$\max(\mathbf{B}^{t+1}) \leq \max(\mathbf{B}^t) - c \cdot \Delta_t. \quad (17)$$

Proof of Claim 2. Since $B_i^t = \max(\mathbf{B}^t)$, every vocal neighbor k satisfies $B_k^t \leq B_i^t$, so $B_k^t - B_i^t \leq 0$ for all $k \in N^t(i)$. With $f_{\alpha_{i,k}} \in \mathcal{F}_\alpha \subset R$: $f_{\alpha_{i,k}}(B_k^t - B_i^t) = \alpha_{i,k}(B_k^t - B_i^t) \leq 0$. Hence all terms in (8) are non-positive, giving $B_i^{t+1} \leq B_i^t$. More precisely, isolating the j -term:

$$\begin{aligned} B_i^{t+1} &= B_i^t + \underbrace{I_{ji} \cdot \alpha_{i,j} \cdot (B_j^t - B_i^t)}_{< 0} + \underbrace{\sum_{k \in N^t(i) \setminus \{j\}} I_{ki} \cdot \alpha_{i,k} \cdot (B_k^t - B_i^t)}_{\leq 0} \\ &\leq B_i^t - I_{ji} \cdot \alpha_{i,j} \cdot (B_i^t - B_j^t) \\ &\leq B_i^t - I_{\min} \cdot \alpha_{\min} \cdot (B_i^t - B_j^t). \end{aligned}$$

Since j can be taken as the min-agent and $B_i^t - B_j^t \geq \Delta_t$: $B_i^{t+1} \leq \max(\mathbf{B}^t) - c \cdot \Delta_t$. Together with $\max(\mathbf{B}^{t+1}) \leq B_i^{t+1}$ (from Lemma 7), Eq. (17) follows. $\square$

Infinitely many contraction steps.

We call step t *productive* if $\Delta_t > 0$ and there exists a vocal neighbor of the max-agent with strictly lower opinion. We show productive steps occur infinitely often.

Suppose for contradiction that after some time T , no step is productive. Then for all $t \geq T$ with $\Delta_t > 0$, every vocal agent at time t holds opinion $\max(\mathbf{B}^t)$. In this situation the min-agent j (with $B_j^t < \max(\mathbf{B}^t)$) is silent at every $t \geq T$. Since j is silent, $S_j^t = 0$; but j still updates its belief:

$$B_j^{t+1} = B_j^t + \sum_{k \in N^t(j)} I_{kj} \cdot \alpha_{j,k} \cdot (B_k^t - B_j^t). \quad (18)$$

Every vocal agent k at time t satisfies $B_k^t = \max(\mathbf{B}^t) \geq B_j^t$, so every term in (18) is non-negative. Hence $\{B_j^t\}_{t \geq T}$ is non-decreasing and bounded above by $\max(\mathbf{B}^T)$. Let $v = \lim_{t \rightarrow \infty} B_j^t \geq B_j^T$.

By Corollary 6, $N^t(j) \neq \emptyset$ for infinitely many t . Whenever $N^t(j) \neq \emptyset$ and $B_j^t < \max(\mathbf{B}^t)$, each vocal k contributes a strictly positive term $I_{kj} \alpha_{j,k} (\max(\mathbf{B}^t) - B_j^t) > 0$. By Lemma 7 and monotonicity (Corollary 8), $\max(\mathbf{B}^t) \geq U > L = \lim \min(\mathbf{B}^t) \geq B_j^t$ (where we use Theorem 9). Hence the increment in (18) is bounded below by $I_{\min} \alpha_{\min} (U - B_j^t) > 0$ at each such step, forcing $v \geq U$, i.e., $B_j^t \rightarrow U$. But B_j^t is a (non-decreasing) sequence bounded above by 1, and $B_j^t \rightarrow U$ means $\min(\mathbf{B}^t) \rightarrow U = \max(\mathbf{B}^t)$: $\Delta_t \rightarrow 0$, contradicting $\Delta_t > 0$ for all $t \geq T$.

Therefore productive steps occur infinitely often.

Geometric bound.

Let $\{t_k\}_{k \geq 0}$ be the (infinite) subsequence of productive steps. By Claim 2 and Corollary 8 (monotonicity of min):

$$\Delta_{t_k+1} \leq \Delta_{t_k} - c \cdot \Delta_{t_k} = (1 - c) \Delta_{t_k}.$$

Since Δ_t is non-increasing between productive steps:

$$\Delta_{t_{k+1}} \leq \Delta_{t_k+1} \leq (1 - c) \Delta_{t_k}.$$

By induction:

$$\Delta_{t_k} \leq (1 - c)^k \Delta_0.$$

For every $\varepsilon > 0$, choose k large enough so that $(1 - c)^k \Delta_0 < \varepsilon$. Then $\Delta_{t_k} < \varepsilon$. $\square$

Theorem 12 (Consensus in CBSOM⁻ cliques with $n \geq 3$). *Let $(G, \mathbf{B}^0, \mathbf{S}^0, \mu_G)$ be a CBSOM⁻ model on an n -agent clique with $n \geq 3$ and $\beta_{i,j} = f_{\alpha_{i,j}} \in \mathcal{F}_\alpha$ for all edges. Then all agents converge to consensus.*

Proof. By Theorem 9, the limits $U = \lim_{t \rightarrow \infty} \max(\mathbf{B}^t)$ and $L = \lim_{t \rightarrow \infty} \min(\mathbf{B}^t)$ exist in $[0, 1]$ with $L \leq U$. We show $U = L$ by contradiction.

Suppose $U - L =: \delta > 0$.

Since $\max(\mathbf{B}^t) \rightarrow U$ and $\min(\mathbf{B}^t) \rightarrow L$, there exists $T_0 \in \mathbb{N}$ such that for all $t \geq T_0$:

$$\max(\mathbf{B}^t) \geq U - \frac{\delta}{4} \quad \text{and} \quad \min(\mathbf{B}^t) \leq L + \frac{\delta}{4}. \quad (19)$$

In particular, for all $t \geq T_0$:

$$\Delta_t = \max(\mathbf{B}^t) - \min(\mathbf{B}^t) \geq \left(U - \frac{\delta}{4}\right) - \left(L + \frac{\delta}{4}\right) = \delta - \frac{\delta}{2} = \frac{\delta}{2} > 0. \quad (20)$$

Thus $\Delta_t \geq \delta/2 > 0$ for all $t \geq T_0$.

Contradiction via Lemma 11.

Applying Lemma 11 with $\varepsilon = \delta/2$ (which is positive), there exists $t^* \in \mathbb{N}$ such that $\Delta_{t^*} < \delta/2$. Taking $t^{**} = \max(T_0, t^*)$ and using the fact that $\{\Delta_t\}$ is non-increasing (Corollary 8):

$$\Delta_{t^{**}} \leq \Delta_{t^*} < \frac{\delta}{2}.$$

But $t^{**} \geq T_0$ implies $\Delta_{t^{**}} \geq \delta/2$ by (20): contradiction.

Conclusion.

Therefore $U = L$. For every agent $i \in A$ and every $t \geq 0$:

$$\min(\mathbf{B}^t) \leq B_i^t \leq \max(\mathbf{B}^t),$$

and both bounds converge to $U = L$. By the Squeeze Theorem [18], $\lim_{t \rightarrow \infty} B_i^t = U = L$ for all $i \in A$. All agents converge to consensus. $\square$